

MYP 1 Design

Handout 6: Exploring PictoBlox Extensions

Learning Objective

By the end of this lesson, you will be able to:

- Explain what an extension is.
- Add an extension in PictoBlox.
- Explore different extensions.
- Create simple programs using extension blocks.

What is an Extension?

An **extension** adds new blocks to PictoBlox. These new blocks give your program **extra features and abilities**. Think of an extension as **adding new tools to your toolbox**.

How to Add an Extension

Follow these steps:

1. Open **PictoBlox**.
2. Click **Add Extension**.
3. Select the extension you want to use.
4. New blocks will appear in the **Block Palette**.

Extension 1: Pen

The **Pen** extension allows the sprite to draw while it moves.

Examples

- Pen Down
- Pen Up
- Clear
- Set Pen Color

Activity

Create this program:

When Green Flag Clicked

↓

Clear

↓

Pen Down

↓

Move 100 Steps

Challenge

Can you make the sprite draw:

- A straight line
- A square
- A triangle

Extension 2: Text to Speech

The **Text to Speech** extension makes the computer speak words aloud.

Examples

- Speak "Hello!"
- Change Voice
- Change Language

Activity

Create this program:

When Green Flag Clicked

↓

Speak

"Welcome to Coding!"

Challenge

Make the computer speak:

- Your name
- Your favourite colour

Extension 3: Translate

The **Translate** extension changes text into another language.

Activity

Translate:

- Hello
- Thank You
- Welcome

Try translating into another language.

Challenge

Can you translate your name or a short greeting?

Extension 4: Music

The **Music** extension lets you play musical notes and sounds.

Examples

- Play Note
- Play Drum
- Set Instrument

Activity

Create a simple tune using three different notes.

Challenge

Can you create your own short melody?

Activity: Combine Extensions

Create a program that:

- Starts with the Green Flag.
- Speaks a welcome message.
- Moves the sprite.
- Draws a line using the Pen extension.
- Plays a musical note.

Explore More

- Choose **one extension** and explore it.
- Complete the table.

Extension	What did you discover?
Pen	
Text to Speech	
Translate	
Music	

Quick Summary

Extension	Purpose
Pen	Draws shapes and lines
Text to Speech	Speaks text aloud
Translate	Changes text into another language
Music	Plays notes and sounds

Mini Challenge

Create your own program using:

- One Event block
- One Motion block
- One Control block
- One Extension

Your program should be creative and demonstrate what you learned.

Reflection

1. What is an extension?
2. Which extension did you enjoy the most?
3. Which extension was the easiest to use?
4. How do extensions make your program more interesting?
5. Which extension would you like to explore in the future?

Homework

Choose **one extension** and create a simple program using it.

Your program should:

- Start with the Green Flag.
- Use at least **one extension block**.
- Include one Motion block.
- Include one Control block.
- Be ready to demonstrate in the next class.